



LATANDROPS 0.005% w/v EYE DROPS

Latanoprost 0.005% w/v

Presentation:

LATANDROPS 0.005% w/v EYE DROPS is a sterile, clear, colourless, odourless, free from visible particles ophthalmic solution. It is formulated as isotonic, preserved ophthalmic solution supplied in a multidose dropper container equipped with a dropper applicator and tamper evident screw cap. Each dropper container contains 2.5mL eye drops solution corresponding to approximately 80 drops of solution.

Composition:

Each mL of eye drops solution contains of active ingredient, latanoprost 50 µg (0.005% w/v). One drop contains approximately latanoprost 1.5 µg. Other excipients include benzalkonium chloride 0.02% w/v as preservative, sodium chloride, sodium dihydrogen phosphate monohydrate, disodium phosphate anhydrous and Water for Injections.

Pharmacodynamic:

The active substance latanoprost, a prostaglandin F_{2a} analogue, is a selective prostanoid FP receptor agonist which reduces the intraocular pressure by increasing the outflow of aqueous humour. Reduction of the intraocular pressure in man starts about three to four hours after administration and maximum effect is reached after eight to twelve hours. Pressure reduction is maintained for at least 24 hours.

Studies in animals and man indicate that the main mechanism of action is increased uveoscleral outflow, although some increase in outflow facility (decrease in outflow resistance) has been reported in man.

Pivotal studies have demonstrated that latanoprost is effective as monotherapy. In addition, clinical trials investigating combination use have been performed. These include studies that show that latanoprost is effective in combination with beta-adrenergic antagonist (timolol). Short term (1 or 2 weeks) studies suggest that the effect of latanoprost is additive in combination with adrenergic agonists (dipivalyl epinephrine), carbonic anhydrase inhibitors (acetazolamide) and at least partly additive with cholinergic agonists (pilocarpine).

Clinical trials have shown that latanoprost has no significant effect on the production of aqueous humour. Latanoprost has not been found to have any effect on the blood-aqueous barrier.

Latanoprost has no or negligible effects on the intraocular blood circulation when used at the clinical dose and studied in monkeys. However, mild to moderate conjunctival or episcleral hyperaemia may occur during topical treatment.

Chronic treatment with latanoprost in monkey eyes, which had undergone extracapsular lens extraction, did not affect the retinal blood vessels as determined by fluorescein angiography.

Latanoprost has not induced fluorescein leakage in the posterior segment of pseudophakic human eyes during short-term treatment.

Latanoprost in clinical doses has not been found to have any significant pharmacological effects on the cardiovascular or respiratory system.

Pharmacokinetics:

Latanoprost (molecular weight 432.58) is an isopropyl ester prodrug which *per se* is inactive but after hydrolysis to the acid of latanoprost becomes biologically active.

The prodrug is well absorbed through the cornea and all drug that enters the aqueous humour is hydrolysed during the passage through the cornea.

Studies in man indicate that the peak concentration in the aqueous humour is reached about 2 hrs after topical administration.

After topical application in monkeys, latanoprost is distributed primarily in the anterior segment, the conjunctiva and the eyelids. Only minute quantities of the drug reach the posterior segment.

There is practically no metabolism of the acid of latanoprost in the eye. The main metabolism occurs in the liver. The half-life in plasma is 17 minutes in man. The main metabolites, the 1,2-dinor and 1,2,3,4-tetranor metabolites, exert no or only weak biological activity in animal studies and are excreted primarily in the urine.

Toxicology:

Preclinical Safety Data: The ocular as well as systemic toxicity of latanoprost has been investigated in several animal species. Generally, latanoprost is well tolerated with a safety margin between clinical ocular dose and systemic toxicity of at least 1000 times. In monkeys, latanoprost has been infused IV in doses of up to 500 µg/kg without major effects on the cardiovascular system. High doses of latanoprost, approximately 100 times the clinical dose/kg body weight, administered IV to unanaesthetized monkeys have been shown to increase the respiration rate probably reflecting bronchoconstriction of short duration. In animal studies, latanoprost has not been found to have sensitising properties.

In the eye, no toxic effects have been detected with doses of up to 100 µg/eye/day in rabbits or monkeys (clinical dose is approximately 1.5 µg/eye/day). In monkeys, however, latanoprost has been shown to induce increased pigmentation of the iris. The mechanism of increased pigmentation seems to be stimulation of melanin production in melanocytes of the iris with no proliferative changes observed. The change in iris colour may be permanent (see also Precautions).

In chronic ocular toxicity studies, administration of latanoprost 6 µg/eye/day has also been shown to induce increased palpebral fissure. This effect is reversible and occurs at doses above the clinical dose level. The effect has not been seen in humans.

Latanoprost was found negative in reverse mutation tests in bacteria, gene mutation in mouse lymphoma and mouse micronucleus test.

Chromosome aberrations were observed *in vitro* with human lymphocytes. Similar effects were observed with prostaglandin F_{2a} , a naturally occurring prostaglandin and indicates that this is a class effect.

Additional mutagenicity studies on *in vitro*/ *in vivo* unscheduled DNA synthesis in rats were negative and indicate that latanoprost does not have mutagenic potency.

Carcinogenicity studies in mice and rats were negative.

Latanoprost has not been found to have any effect on male or female fertility in animal studies. In the embryotoxicity study in rats, no embryotoxicity was observed at IV doses (5, 50 and 250 µg/kg/day) of latanoprost. However, latanoprost induced embryo-lethal effects in rabbits in doses ≥ 5 µg/kg/day.

The dose of 5 µg/kg/day (approximately 100 times the clinical dose) caused significant embryo-fetal toxicity characterised by increased incidence of late resorption and abortion and by reduced foetal weight.

No teratogenic potential has been detected.

Indications:

Reduction of elevated intraocular pressure in patients with open-angle glaucoma, chronic angle closure glaucoma (CACG) and ocular hypertension.

Dosage and Administration:

Adults (Including the Elderly):

Recommended dosage is 1 eye drop in the affected eye(s) once daily. Optimal effect is obtained when **LATANDROPS** is administered in the evening.

The dosage of **LATANDROPS** should not exceed once daily since it has been shown that more frequent administration decreases the intraocular pressure-lowering effect.

If one dose is missed, treatment should continue with the next dose as normal.

As with any other eye drops, to reduce possible systemic absorption, it is recommended that the lacrimal sac be compressed at the medial canthus (punctal occlusion) for one minutes. This should be performed immediately following the instillation of each drop.

Contact lenses should be removed before instillation of the eye drops and may be reinserted after 15 minutes.

If more than one topical ophthalmic drug is being used, the drugs should be administered at least five minutes apart.

Use in Children:

Safety and effectiveness in children has not been established. Therefore, **LATANDROPS** is not recommended for use in children.

Contraindications:

Known hypersensitivity to any component of **LATANDROPS**.





Special Warning and Precautions:

LATANDROPS may gradually change the eye colour by increasing the amount of brown pigment in the iris. This effect has predominantly been seen in patients with mixed coloured irides i.e. blue-brown, grey-brown, green-brown or yellow-brown, and is due to increased melanin content in the stromal melanocytes of the iris. Typically, the brown pigmentation around the pupil spreads concentrically towards the periphery in the affected eyes, but the entire iris or parts of it may become more brownish. In patients with homogeneously blue, grey, green or brown eyes, the change has only rarely been seen during 2 years of treatment in clinical trials. The change in iris colour occurs slowly and may not be noticeable for several months to years. It has not been associated with any symptom or pathological changes in clinical trials.

No further increase in brown iris pigment has been observed after discontinuation of treatment, but the resultant colour change may be permanent.

Until further long-term data is obtained, it is recommended that in patients with mixed coloured irides, only those who are intolerant or insufficiently responsive to another intraocular pressure-lowering medication be treated.

Neither naevi nor freckles of the iris have been affected by treatment. Accumulation of pigment in the trabecular meshwork or elsewhere in the anterior chamber has not been observed in clinical trials, but until further long-term experience about increased iris pigmentation is available, patients should be examined regularly and, depending on the clinical situation, treatment may be stopped if increased iris pigmentation ensues.

Before treatment is instituted, patients should be informed of the possibility of a change in eye colour. Unilateral treatment can result in permanent heterochromia.

LATANDROPS has no or little effect on the pupil, but there is no experience in acute attacks of closed-angle glaucoma. Therefore, it is recommended that **LATANDROPS** should be used with caution in these conditions until more experience is obtained.

LATANDROPS contains benzalkonium chloride which may be absorbed by contact lenses. The contact lenses should be removed before instillation of the eye drops and may be reinserted after 15 min.

Use **LATANDROPS** with caution in patients with asthma.

Effects on the Ability to Drive or Operate Machinery: In common with other eye preparations, instillation of eye drops may cause transient blurring of vision. Wait until this wears off before driving or operating machinery.

Use in Pregnancy & Lactation:

LATANDROPS does not increase the spontaneous incidence of birth defects, but it has potential hazardous pharmacological effects with respect to the course of pregnancy, to the unborn or the neonate.

Latanoprost has been shown to cause embryofoetal toxicity in rabbits characterised by increased incidences of late resorption and abortion and reduced foetal weight when given in IV doses approximately 100 times the human dose.

LATANDROPS should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Latanoprost and its metabolites may pass into breast milk and **LATANDROPS** should therefore be used with caution in nursing women.

Drug Interactions:

The intraocular pressure-reducing effect of latanoprost has been shown to be additive to that of β -adrenergic antagonist (timolol), adrenergic agonists (dipivalyl epinephrine), carbonic anhydrase inhibitors (acetazolamide), and at least partly to cholinergic agonists (pilocarpine).

Incompatibilities: *In vitro* studies have shown that precipitation occurs when eye drops containing thiomersal are mixed with **LATANDROPS**.

If such drugs are used, the eye drops should be administered with an interval of at least 5 min.

Adverse Reactions:

LATANDROPS has caused increased pigmentation of the iris (see also Precautions).

LATANDROPS may cause darkening, thickening and lengthening of eye lashes. Darkening of the palpebral skin of the lids has very rarely been noted.

LATANDROPS may cause a slight foreign body sensation. Mild conjunctival hyperaemia has been noted in about 10% and moderate hyperaemia in about 1% of patients undergoing chronic treatment. Transient punctate epithelial erosions, mostly without symptoms, have been noted in some patients.

Macular oedema has been reported rarely during **LATANDROPS** treatment. These reports have mainly occurred in patients with risk factors for macular oedema. Rare cases of iritis/uveitis have been reported.

Rare cases of asthma, asthma aggravation, acute asthma attack and dyspnoea have been reported. There is limited experience from patients with asthma but latanoprost has not been found to affect the pulmonary function when studied in a small number of steroid-treated patients suffering from moderate asthma, neither has it been found to affect the pulmonary function, airway reactivity or β_2 -responsiveness when studied in a small number of nonsteroid-treated asthma patients.

Skin rash of unknown etiology has been rarely reported during **LATANDROPS** treatment.

LATANDROPS is usually well-tolerated, but a slight redness in the patient's eye(s) may occur. The patient might feel as if there is something in the eye(s). A few patients have noticed changes in their vision that did not always occur, just after administering the drops. These changes may mean that reading and being able to see fine detail is difficult. On the other hand, vision may become blurred and eye(s) look red and feel sore/painful. Although unlikely, if either happens, discontinue use and contact the physician.

Overdosage:

Apart from ocular irritation and conjunctival hyperaemia, no other ocular side effects are known if **LATANDROPS** is overdosed.

If **LATANDROPS** is accidentally ingested, the following information may be useful: One bottle contains latanoprost 125 μ g. More than 90% is metabolised during the first-pass through the liver. IV infusion of 3 μ g/kg in healthy volunteers induced no symptoms but a dose of 5.5-10 μ g/kg caused nausea, abdominal pain, dizziness, fatigue, hot flushes and sweating. When latanoprost was applied in a dose 7 times the clinical **LATANDROPS** dose in patients with moderate bronchial asthma, no significant effects were seen on respiratory function, heart rate or blood pressure.

If overdosage with **LATANDROPS** occurs, treatment should be symptomatic.

Storage Conditions:

Store in a refrigerator (2°C - 8°C).

Keep the bottle in the outer carton in order to protect from light.

Once opened, the container should be used within 4 weeks and may be stored at room temperature up to 25°C.

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